Lecture 30

Turbulence

- cascade
- Log law

Last time : considered averaging gov. eq to find $\bar{u}$, which is a "goal"

$$\underset{\sim}{u} = \underset{\sim}{\bar{u}} + \underset{\sim}{u}'$$

mean

- linear eq $\rightarrow$ no problem (mean is a linear operation)
- non linear eq. (e.g. N-S) $\rightarrow$ closure problem

$\underset{\sim}{\bar{u}}$ and $\overline{\underset{\sim}{u}\underset{\sim}{u}}$ are unknowns

average NS (incompressible)

NOTE: no random ξ term:

$\rightarrow$ chaotic dynamics that we'd like to average over

$$\frac{\partial u_i}{\partial t} + \underbrace{u_j \frac{\partial u_i}{\partial x_j}} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j \partial x_j}$$

$$\frac{\partial u_i u_j}{\partial x_j} = u_i \cancel{\frac{\partial u_j}{\partial x_j}} + u_j \frac{\partial u_i}{\partial x_j}$$

$\nabla \cdot \underset{\sim}{u} = 0$ incompressible

$$u_i(\underset{\sim}{x}, t) = \bar{u}_i(\underset{\sim}{x}) + u_i'(\underset{\sim}{x}, t)$$

mean

fluctuation

$\overline{u'} = 0$

ave both sides

$$\overline{u_i} = \overline{\bar{u}_i + u_i'}$$

$\bar{\bar{u}}_i = \bar{u}_i$

$\overline{u'} = 0$

$$\frac{\partial \overline{(\bar{u}_i + \cancel{u_i'})}}{\partial t} + \frac{\partial \overline{(\bar{u}_i + u_i')(\bar{u}_j + u_j')}}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \overline{(\bar{P} + \cancel{p'})}}{\partial x_j} + \nu \frac{\partial^2 \overline{(\bar{u}_i + \cancel{u_i'})}}{\partial x_j \partial x_j}$$

$$\frac{\partial \overline{\bar{u}_i \bar{u}_j}}{\partial x_j} + \frac{\partial \overline{\bar{u}_i u_j'}}{\partial x_j} + \frac{\partial \overline{u_i' \bar{u}_j}}{\partial x_j} + \frac{\partial \overline{u_i' u_j'}}{\partial x_j}$$

$$\frac{\partial \bar{u}_i \bar{u}_j}{\partial x_j} + \emptyset + \emptyset + \frac{\partial \overline{u_i' u_j'}}{\partial x_j}$$

$$\overline{\nabla \cdot \underline{u}} = \nabla \cdot \overline{(\underline{\bar{u}} + \underline{u}')} = \nabla \cdot \underline{\bar{u}}$$

$$\bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j}$$

viscous stress on $\bar{u}_i$

"Reynolds stress"

Reynolds Average N-S

$$\bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_j} + \underbrace{\frac{\partial}{\partial x_j}\left(\nu \frac{\partial \bar{u}_i}{\partial x_j} - \overline{u_i' u_j'}\right)}_{\nabla \cdot \underline{\underline{\tau}}}$$

viscous + turbulence stresses

BL = ?

$i = 1$

x-mom

$$\bar{u}\frac{\partial \bar{u}}{\partial x} + \bar{v}\frac{\partial \bar{u}}{\partial y} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x} + \nu \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial \overline{u'v'}}{\partial y}$$

unclosed

$\frac{\partial}{\partial y}$ large

$\bar{v}$ small

turbulence modeling : express $\overline{u_i' u_j'}$ in terms of $\overline{u}_i$

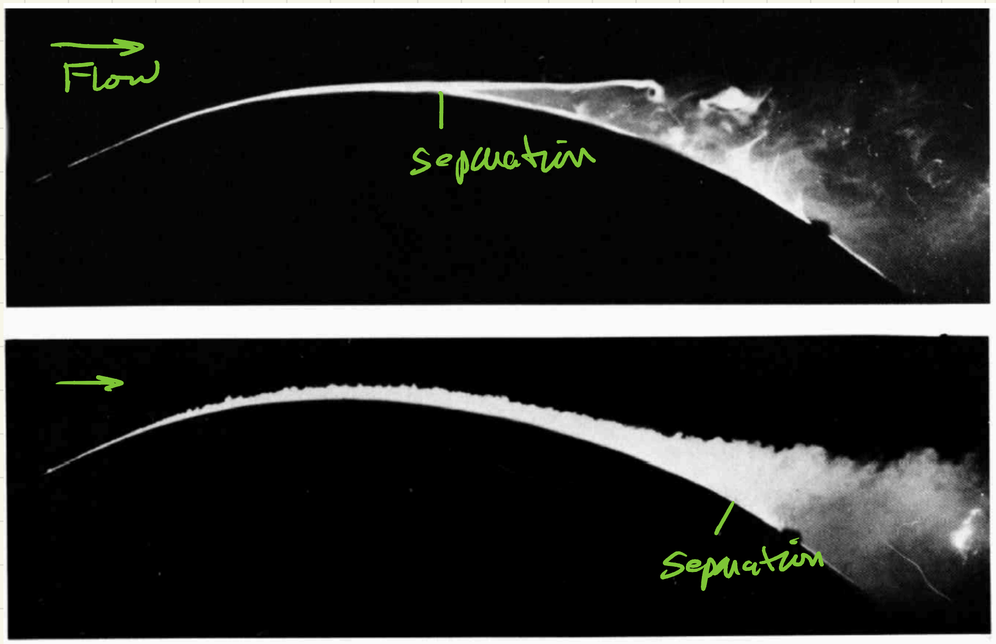
Kinda, sorta works

→ good where models are tuned

→ less good away from "calibration" flow

→ poor for separation

Extra Reynolds stress keeps flows "attached"



lower Re, laminar

turbulent

156. Comparison of laminar and turbulent boundary layers. The laminar boundary layer in the upper photograph separates from the crest of a convex surface (cf. figure 38), whereas the turbulent layer in the second photograph remains attached; similar behavior is shown below for a sharp corner. (Cf. figures 55-58 for a sphere.) Titanium tetrachloride is painted on the forepart of the model in a wind tunnel. *Head 1982*

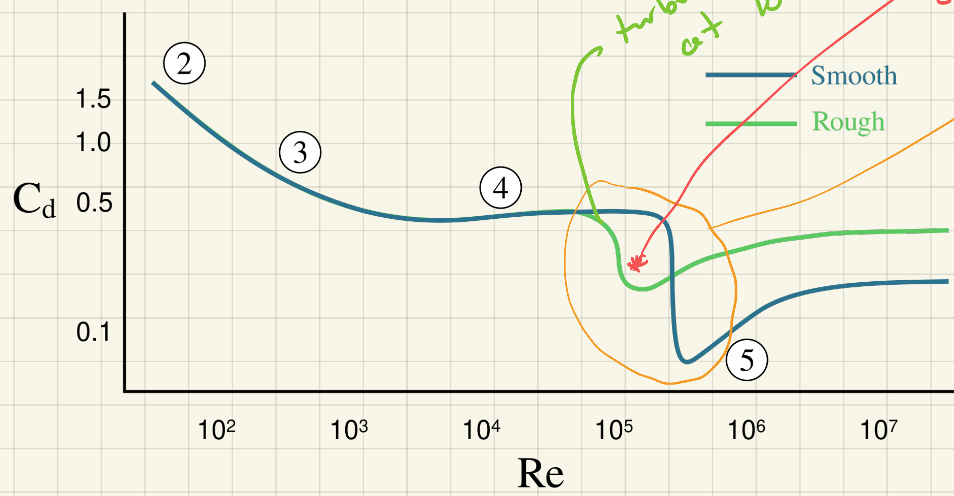
Circle in cross flow

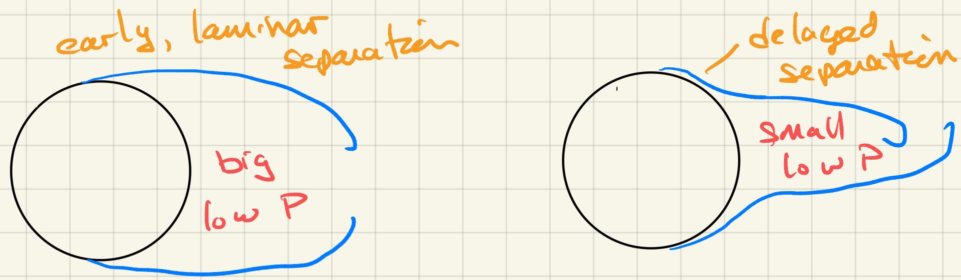
turbulent at lower Re

golf ball ??

drag crisis

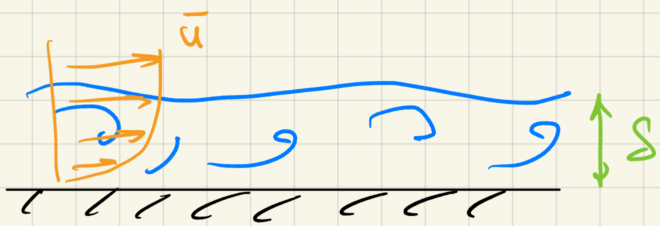
Drag





Both Smooth
(or both rough)

Turbulence Scales → Energy Cascade



- largest turbulent scale $\approx \delta$

  - $Re = \dfrac{\delta U}{\nu} >> 1$ — not a viscous BL — turbulence stress BL

- smallest turbulent scale $\approx \eta$ (Kolmogorov)

  - acted on by viscosity $Re_\eta = \dfrac{u_\eta \eta}{\nu} \approx 1$

Kolmogoroff : noted scale mismatch

conjectured scales only interact with similar scales

reasoning based also on wave # (inverse lengths)

$k_1, k_2 \longrightarrow$ quadratic nonlinearity

$e^{ik_1x} e^{ik_2x} \longrightarrow \pm(k_1+k_2)$

$\pm(k_1-k_2)$

Turbulence Kinetic Energy Cascade

inviscid

mean flow $\xrightarrow{\varepsilon}$ large "eddies"

$\xrightarrow{\varepsilon}$ smaller "eddies" still inviscid

$\xrightarrow{\varepsilon}$ smallest eddies viscous

$\xrightarrow{\varepsilon}$ heat

equilibrium: energy in $\approx$ energy out

- ε viscous dissipation rate $\rightarrow$ rate of energy passing through all the scales
  - dimensions?

$$KE \doteq \frac{ML^2}{T^2}$$

$$\varepsilon = \frac{\overset{\frac{ML^2}{T^2}}{KE}}{\underset{\sim M}{\text{mass}}} \times \overset{1/T}{\text{rate}} \doteq \frac{L^2}{T^3}$$

- consider energy at scale r

- $r \ll L$ or δ away from largest scales

- $r \gg \eta$ away from viscous scale

ask: how much TKE associated with r?

$$E(r) = f(\cancel{\delta}, \cancel{\nu}, \varepsilon, r)$$

energy / mass

L^2 / T^2

inertial scales

nondimensionalize ... 1 Π group

$$\frac{\overset{L^2/T^2}{E(r)}}{\underbrace{(\varepsilon r)}_{L^3/T^3}{}^{2/3}} = \text{const}$$

$(\varepsilon r)^{2/3} \sim (\varepsilon/k)^{2/3}$

$k \sim 1/r$

$$E(r) = \int E(k)\, e^{ikr}\, dk$$

$\sim (\varepsilon/k)^{2/3}$

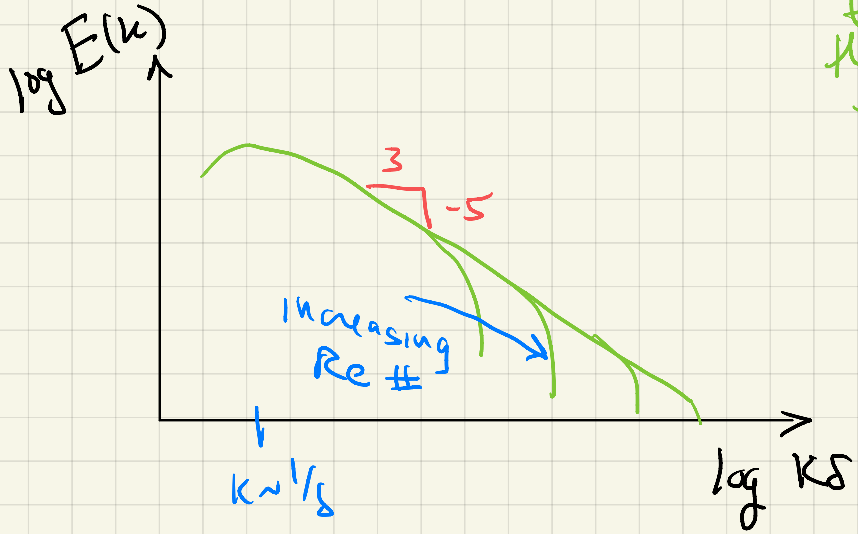
$\sim k$

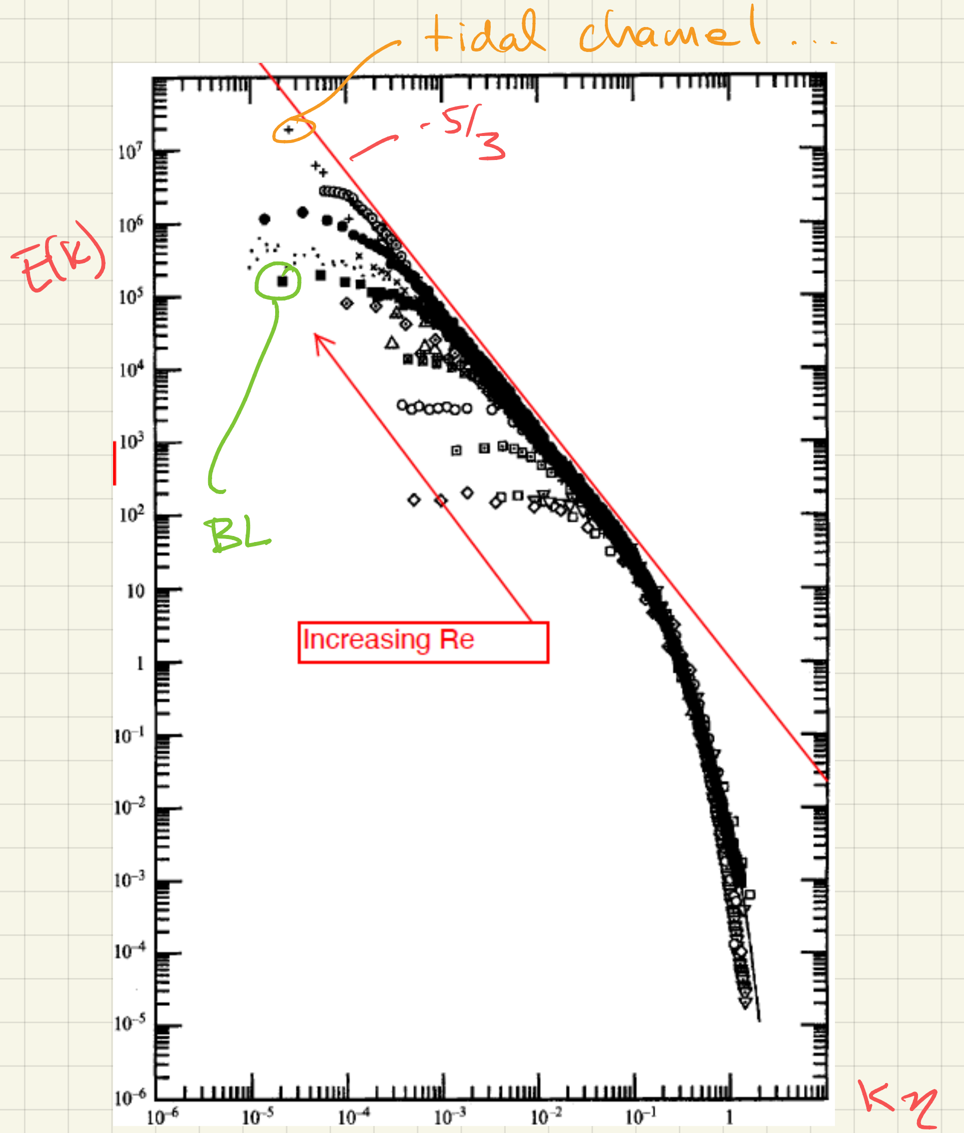
$$\left(\frac{\varepsilon^{2/3}}{k^{5/3}}\right)$$

$$\boxed{E(k) \sim \varepsilon^{2/3}\, k^{-5/3}}$$

const for that flow

shape of the "inertial range" of the spectrum





Saddoughi JFM 1994